



DIGITAL COMMUNICATION LABORATORY

Experiment 5

DPCM and Delta Modulation

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Software: Python / Google Colab

Python • Simulation • Distortion Analysis • Step Validation

Objectives

- To implement a **first-order DPCM predictor**.
- To reduce sample redundancy using **prediction**.
- To compare **PCM and DPCM quantization errors**.
- To implement **Delta Modulation (DM)** for different step sizes.
- To study the effect of **slowly and rapidly varying input signals**.
- To identify **granular noise** and **slope-overload distortion**.
- To verify that every DM staircase step changes by exactly **$+\Delta$ or $-\Delta$** .

Why Prediction?

- PCM codes every sample independently — ignores correlation between neighbours
- Speech/audio samples are highly correlated: current sample $\approx$ previous sample
- DPCM idea: transmit only the prediction error $e(n) = x(n) - \hat{x}(n)$
- Error has much smaller dynamic range $\rightarrow$ same SNR with fewer bits

First-order predictor: $\hat{x}(n) = \alpha \cdot \tilde{x}(n-1)$, typically $\alpha \approx 1$

- At the receiver the predictor is rebuilt from decoded samples $\rightarrow$ lossless structure preserved

First-Order DPCM — Signal Flow

1. Predict

$\hat{x}(n) = \alpha \cdot \hat{x}(n-1)$ — form estimate from past reconstructed sample

2. Subtract

$e(n) = x(n) - \hat{x}(n)$ — small-amplitude prediction error

3. Quantize

$\hat{e}(n) = Q[e(n)]$ — error coded with N bits (fewer than PCM)

4. Reconstruct

$\tilde{x}(n) = \hat{x}(n) + \hat{e}(n)$ — fed back to the predictor (closed loop)

PCM vs DPCM — Comparison

Aspect	PCM	First-order DPCM
What is coded	Each raw sample $x(n)$	Prediction error $e(n) = x(n) - \hat{x}(n)$
Dynamic range	Full signal range	Much smaller $\rightarrow$ finer quantizer steps
Bits per sample	N bits (e.g. 8)	$N - 2 \dots N - 4$ bits for equal SNR
Error behaviour	Quantization noise only	Prediction error + quantizer noise (feedback keeps it bounded)

Delta Modulation — 1-Bit DPCM

Delta Modulation uses only **one bit per sample** to indicate whether the reconstructed signal should move **up or down**.

Decision

If $x[n] \geq \hat{x}[n-1]$
then: $\hat{x}[n] = \hat{x}[n-1] + \Delta$
Otherwise: $\hat{x}[n] = \hat{x}[n-1] - \Delta$

Therefore:

Bit 1 $\rightarrow +\Delta$

Bit 0 $\rightarrow -\Delta$

Important parameter

$\Delta = \text{Step Size}$

The step size determines how quickly the staircase can follow the input.

Simulation Setup

Parameter	Value
Sampling frequency	1000 Hz
Signal duration	1 s
Slowly varying frequency	5 Hz
Rapidly varying frequency	80 Hz
DPCM quantization step	0.1
DM small step	0.03
DM moderate step	0.10
DM large step	0.30

Signals used

Slow signal:

$$x_s(t) = \sin(2\pi 5t)$$

Rapid signal:

$$x_f(t) = \sin(2\pi 80t)$$

Code link

- **Google Colab link:**

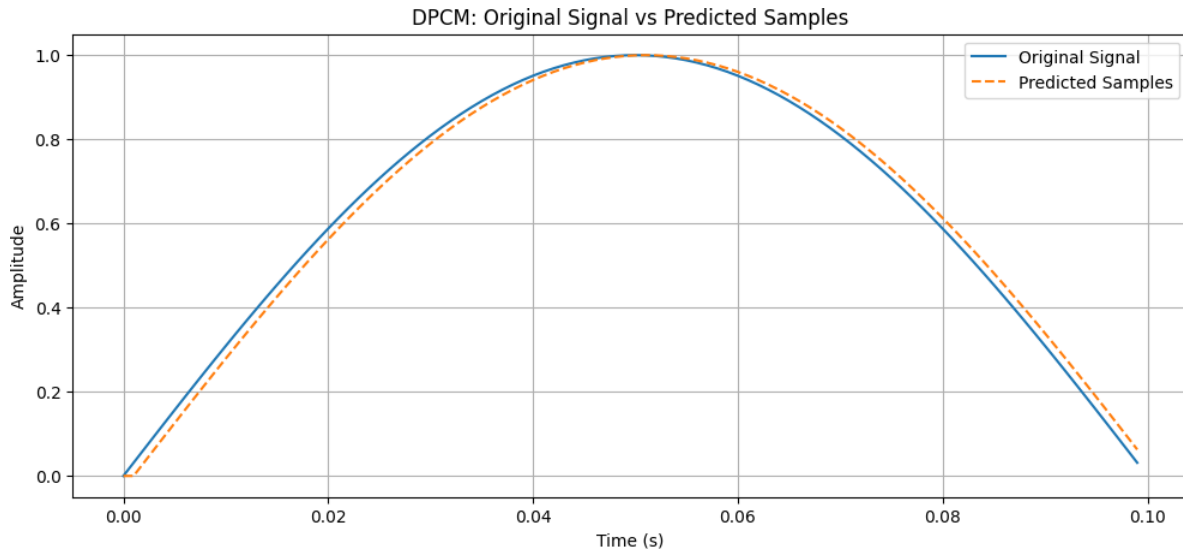
<https://colab.research.google.com/drive/1a5QHwk1ELh-w0DeI2mVk8lhflLeB4dvQJ?usp=sharing>

- **Git hub link:**

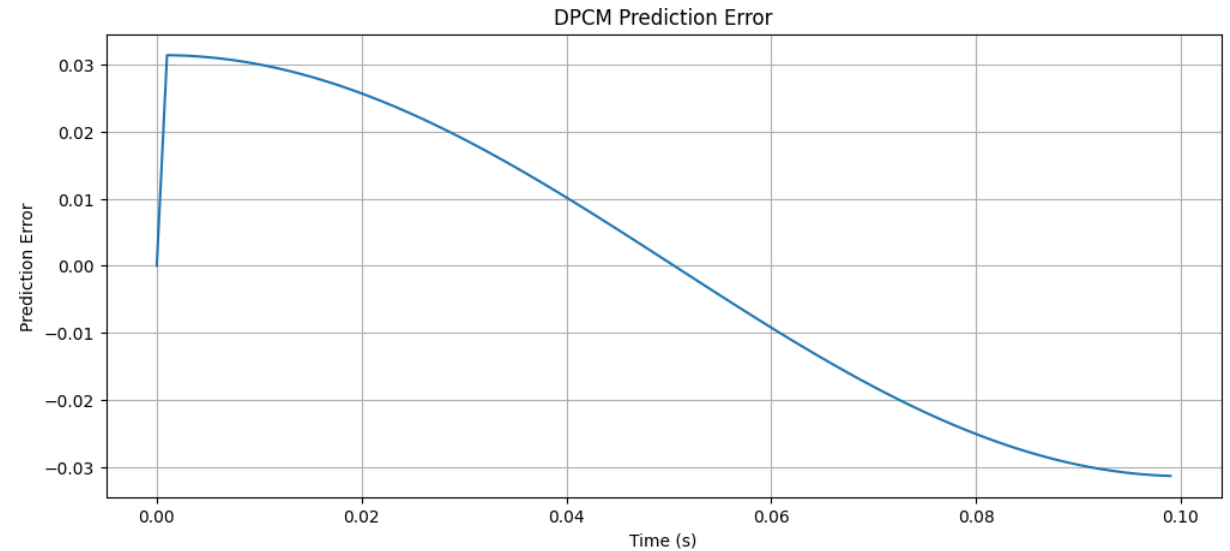
<https://github.com/royayan959/DPCM-and-Delta-Modulation>

DPCM Results

GRAPH-1



GRAPH-2

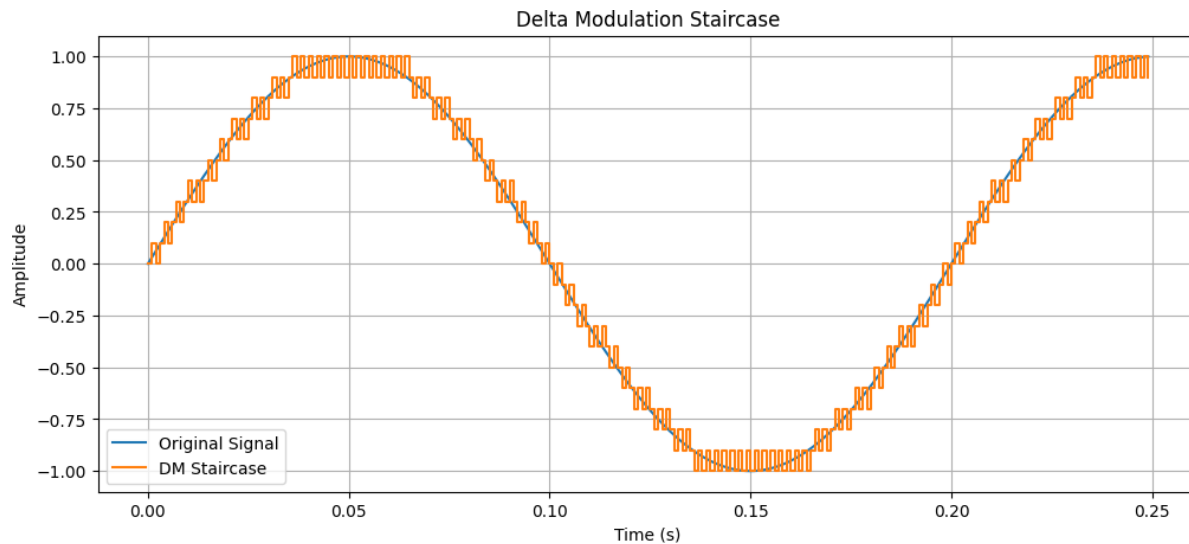


Results:

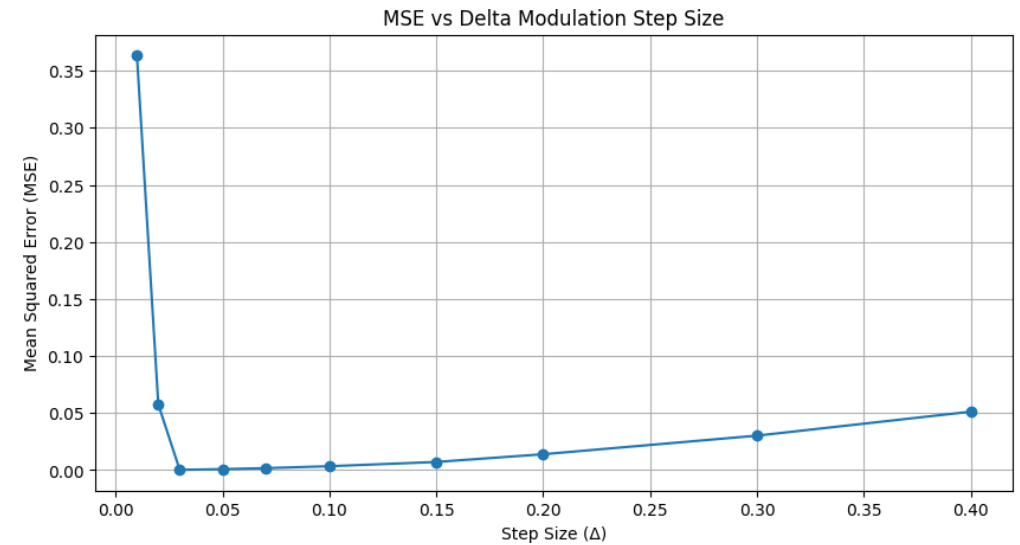
- The predicted signal follows the original signal closely.
- The prediction error is much smaller than the original signal amplitude.
- DPCM represents the **difference signal** rather than the complete signal.

Delta Modulation Results

GRAPH-3



GRAPH-4



Observation:

- **Small Δ :** staircase changes slowly and may fail to follow rapid signal changes.
- **Moderate Δ :** better tracking of the input.
- **Large Δ :** larger jumps can cause oscillations around the input.

Granular Noise vs Slope Overload

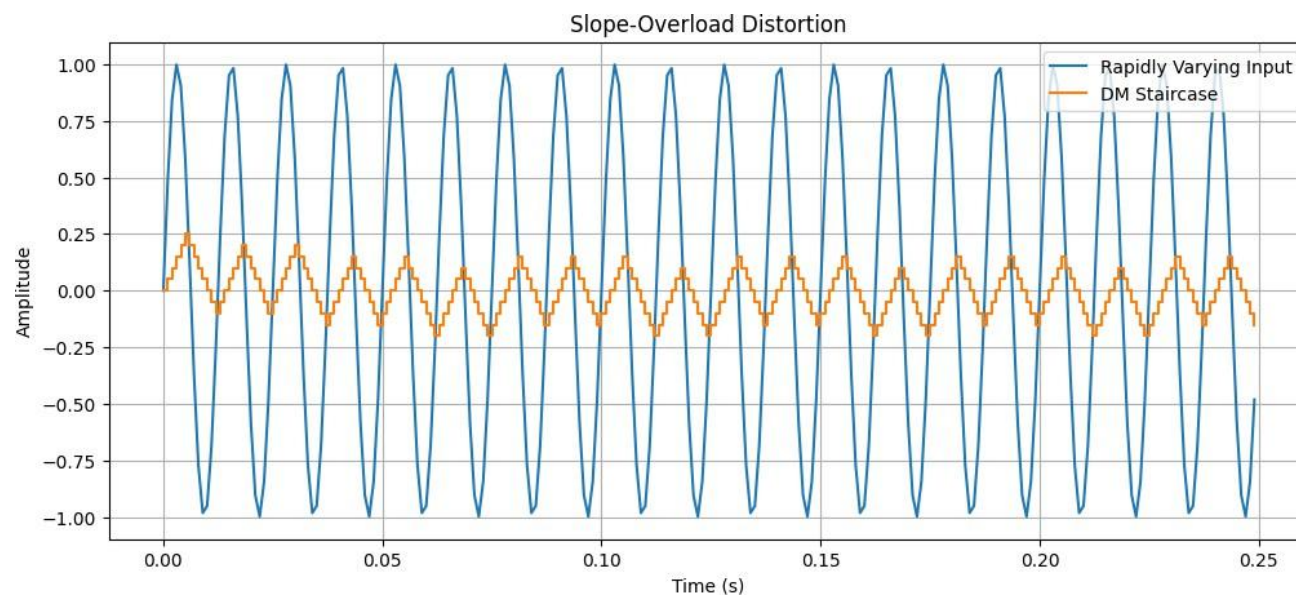
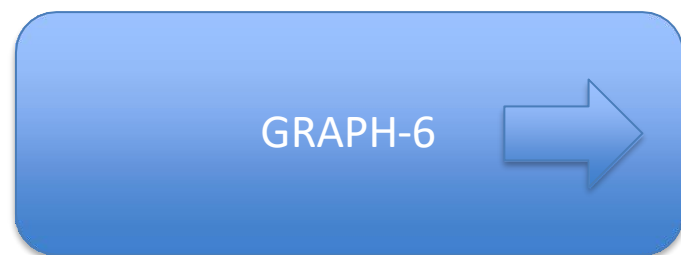
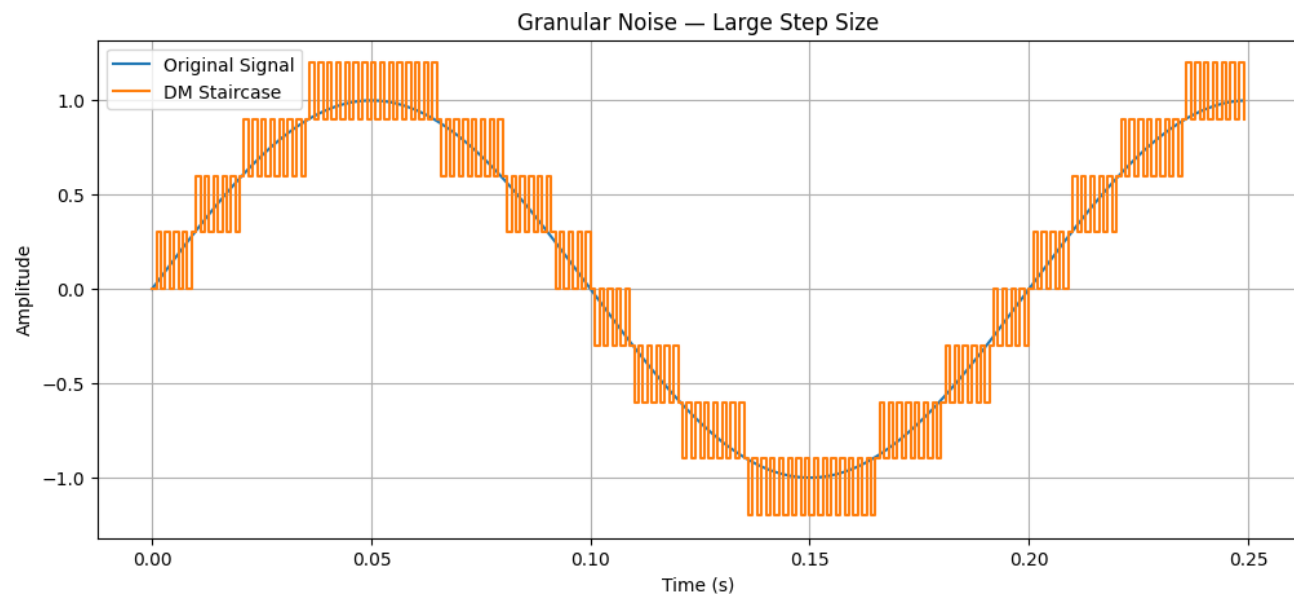
Granular noise

- Cause: step Δ too large relative to slow input changes
- Staircase hunts around a nearly-flat signal — continuous $\pm\Delta$ ripple
- Dominant for small input slope $|dx/dt| \ll \Delta \cdot f_s$

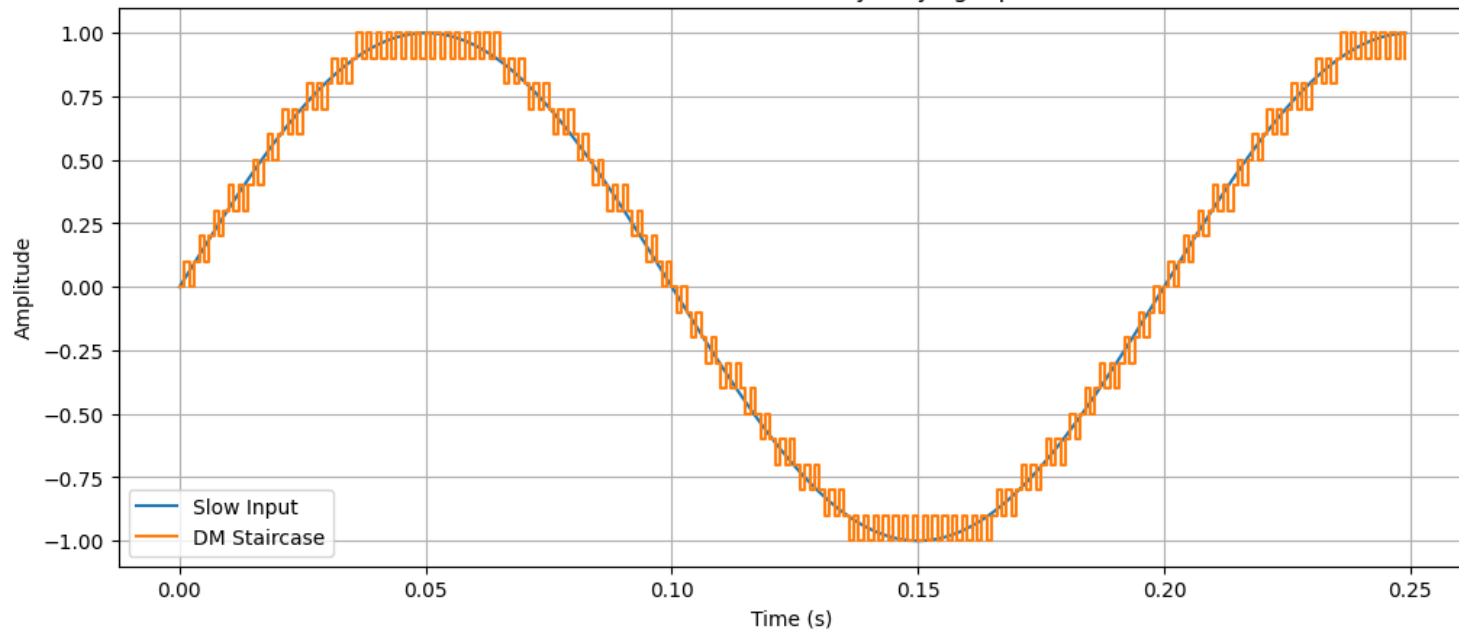
Slope-overload distortion

- Cause: step Δ too small for a fast input
- Max staircase slope = $\Delta \cdot f_s < \text{required slope}$ $\rightarrow$ staircase lags the waveform
- Looks like clipping of peaks; grows with input frequency

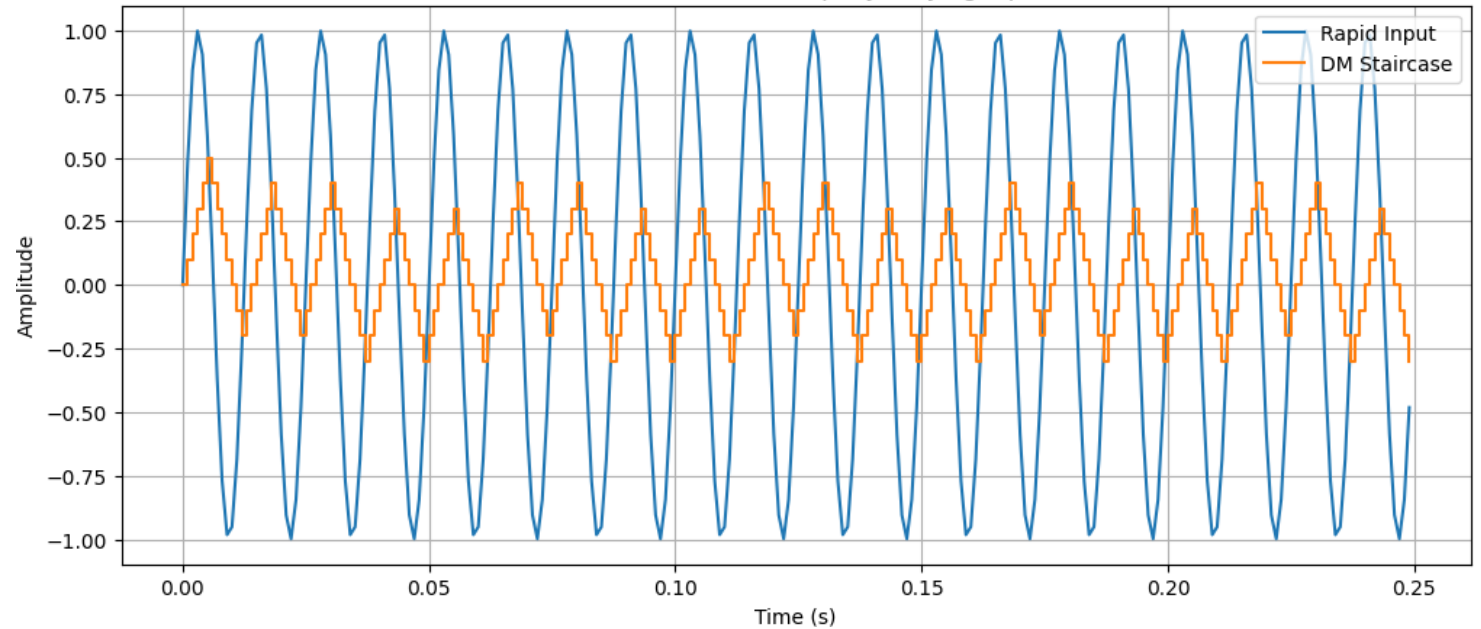
Rule of thumb: avoid overload with $\Delta \cdot f_s \geq \max |dx/dt|$, then keep Δ as small as possible



Delta Modulation — Slowly Varying Input



Delta Modulation — Rapidly Varying Input



Observation & Interpretation Protocol

Validation

The Delta Modulation (DM) reconstructed signal must satisfy:

$$\hat{x}[n] - \hat{x}[n-1] = +\Delta \quad \text{or} \quad -\Delta$$

- Every output step changes strictly by $+\Delta$ or $-\Delta$.

Overall Observations

- **Small Step (Δ):** Fine resolution, but prone to *slope-overload distortion* on rapid changes.
- **Large Step (Δ):** Fast response, but causes *granular noise* on slowly changing signals.
- **Signal Frequency:** High-frequency inputs require a larger Δ to avoid lag.

Conclusions

- DPCM reduces redundancy by encoding the **prediction error**.
- The first-order predictor successfully follows the input signal.
- Delta Modulation uses **+ Δ and $-\Delta$** steps to reconstruct the signal.
- Step size strongly affects reconstruction quality.
- **Granular noise** occurs with an excessively large step size.
- **Slope overload** occurs when the step size is too small for a rapidly varying input.
- Proper selection of Δ provides a balance between tracking ability and quantization noise.